Zhao Zhang

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EDUCATION

Osaka University (OU) Osaka, Japan
Doctor Candidate: Astrophysics *10/2022-*
Degree: Doctor of Science (PhD)
Major & Research Area: Astrophysics and Cosmology
Research Focus: Probing cosmic baryons with FRBs based on cosmological hydrodynamic simulations.
Research Description: Used FRB dispersion measures and SPH-based simulations (CROCODILE) to study cosmic baryon distribution and AGN feedback. Recently expanded the research to explore how SNR evolution affects FRB DM source contributions through 1D numerical modeling.

Nanjing University (NJU) Nanjing, China
Master of Science: Astrophysics *09/2019-2022*
Degree: Master of Science (MSc)
Major & Research Area: Astrophysics
Thesis Title: The Physical Origins and Radiation Mechanisms of High-Energy Transient Events in Magnetars.
Thesis Description: Studied various models of magnetar-powered high-energy transients, including giant flares and X-ray bursts associated with FRBs. Developed emission models and performed spectral fitting to interpret observational data from GRB-like events and magnetar-asteroid interactions.

Guangxi University (GXU) Nanning, China
Bachelor of Science: Physics, Astrophysics *09/2015-07/2019*
Degree: Bachelor of Science (BSc)
Major & Research Area: Physics
Thesis Title: Intrinsic Spectral Shape of Gamma-ray Bursts: A Case Study of GRB 170405A
Thesis Description: Investigated the spectral characteristics of GRB170405A, focusing on an anomalous low-energy bump in its prompt emission. Applied multi-model spectral fitting using McSpecFit to distinguish thermal and non-thermal components and discussed the burst's intrinsic synchrotron origin.

Ruiquan High School Weinan, China
Address: 132 Laocheng St, Linwei District, Weinan, Shaanxi, China 09/2012-07/2015
Focus: General Secondary Education

RESEARCH EXPERIENCE

Modeling FRB propagation through supernova remnants using hydrodynamic simulations *03/2025-*
– Using one/three-dimensional hydrodynamic simulation to Investigate the free-free absorption and FRB dispersion measure (DM) contributions from the SNR environment, focusing on the temporal variation of $\mathrm{DM}_{\mathrm{source}}$ to constrain progenitor properties and remnant structures.
Advisor: Prof. Nagamine and Prof. Herman

Probing cosmic baryons with FRBs based on cosmological hydrodynamic simulations. *10/2022-03/2025*
-- Using fast radio bursts (FRBs) and Smoothed Particle Hydrodynamics (SPH) simulations to investigate the missing baryon problem, analyzing dispersion measures to elucidate the universe's baryonic distribution and evolution.
Advisor: Prof. Nagamine

A Comptonized Fireball Bubble: Physical Origin of Magnetar Giant Flares *10/2020-10/2021*
-- Proposing a new readily fit-able model for the MGFs from the first principle, which invokes an expanding fireball Comptonized by the relativistic magnetar wind at photosphere radius.
Advisor: Prof. Bin-Bin Zhang

A hyper flare of a weeks-old magnetar born from a binary-neutron-star merger *10/2021-01/2022*
-- Reporting the identification of a magnetar that was only a few weeks old, which bridges the gap between the presumed millisecond magnetars and the elderly magnetars observed in the Milky Way Galaxy, indicating a broader channel of magnetar-powered gamma-ray transients.
Advisor: Prof. Bin-Bin Zhang

X-ray burst associated with FRB 200428 by the from magnetar-asteroid impact *03/2021-10/2021*
-- Proposing that *XRB associated with FRB 200428 could be generated by synchrotron emission from the internal shocks between fast and slow shells formed by a relativistically expanding fireball.*
Advisor: Prof. Zi-Gao Dai

A Direct Fit to a Large Sample of GRB Spectra with Synchrotron Radiation Model *10/2019-10/2020*
-- Based on parallel numerical calculations of GPU, solving the electron continuity equation and reproduce the synchrotron radiation spectrum.
Advisor: Prof. Bin-Bin Zhang

Gravitational Lensing for Two GRB Events *10/2020-10/2021*
-- Based on the theory of gravitational lensing, performing analysis for two GRBs, whose coordinates and spectral shape are very consistent with each other.
Advisor: Prof. Bin-Bin Zhang

GRB Analysis Pipeline for GECAM and HXMT Mission *06/2021-10/2020*
-- Based on python, developing the data processing pipeline for high-energy data observed by GECAM and HXMT satellite.
Advisor: Prof. Bin-Bin Zhang

Intrinsic Spectral Shape of Gamma-ray Bursts: A Case Study of GRB 170405A *10/2015-09/2016*
-- Based on the time-resolved spectral analysis to explore the nature of the thermal-like component appearing in the GRB photon spectrum.
Advisor: Prof. Hou-Jun Lü and Prof. Bin-Bin Zhang

PUBLICATIONS

First-Authored Papers

- **Zhao Joseph Zhang,** Gaku Kawashima, Herman Lee, Kentaro Nagamine, Bing Zhang et al., Probing the Dispersion and Rotation Measure Contributions from Supernova Remnants in Fast Radio Burst Source Environments with 1D SNR Simulation, *ApJ submitted* (*arXiv:2603.07012*) (2026)
- **Zhao Joseph Zhang,** Kentaro Nagamine, Oku Yuri, K.G. Lee, Bing Zhang et al., Probing the cosmic baryon distribution and the impact of AGN feedback with FRBs in CROCODILE simulation, *ApJ*, 993, 162 (2025)
- **Zhao Joseph Zhang**, Bin-Bin Zhang and Yan-Zhi Meng, Comptonized Fireball Bubble: Physical Origin of Magnetar Giant Flares, *MNRAS*, 520, 6195 (2023)

Co-Authored Papers

- Umer Rehman, **Zhao Joseph Zhang**, Bin-Bin Zhang, A First-Principles Formulation of High-Energy Radiation from Magnetar Giant Flares, *Journal of High Energy Astrophysics*, 50, 100489 (2026).
- Khrykin, I. S., Tejos, N., Prochaska, J. X., Mannings, A., Mas-Ribas, L., Nagamine, K., Lee, K.-G., Gaensler, B. M**., Zhang, Zhao Joseph**, et al.， Magnetic fields in galactic environments probed by fast radio bursts, *Astronomy & Astrophysics*, 706, A11. (2026)
- Ying-Ze Shan, Xing Yang, Hou-Jun Lü, Jared Rice, Hao-Yu Yuan; Xue-Zhao Chang**, Zhao Joseph Zhang** et al., Signature of a magnetar central engine with precession motion in the X-ray emission of GRB 220711B, *MNRAS*, 541, 3787 (2025)
- Yi-Han Iris Yin, **Zhao Joseph Zhang** et al., A Comptonized Fireball Bubble Fits the Second Extragalactic Magnetar Giant Flare GRB 231115A, *ApJL*, 961, L10 (2024)
- Ze-Nan Liu, Yuan-Pei Yang, Wei-Yang Wang, **Zhao Joseph Zhang** et al., An X-Ray Burst Associated with FRB 200428 from a Magnetar-asteroid Impact, *Research in Astronomy and Astrophysics***,** 23, 8 (2023)
- Bin-Bin Zhang, **Zhao Joseph Zhang** et al., A hyper flare of a weeks-old magnetar born from a binary-neutron-star merger *arXiv*: 2205.07670 (2022)
- Dao-Zhou Wang, Xiao-Hong Zhao, **Zhao Joseph Zhang**, et al., A Comprehensive Consistency Check between Synchrotron Radiation and the Observed Gamma-Ray Burst Spectra, *ApJ*, 178, 12 pp (2021)
- Xing Yang, Hou-Jun Lü, Hao-Yu Yuan, **Zhao Zhang** et al., Evidence for gravitational lensing of GRB 200716C*, ApJL*, 921, L29 (2021).
- Gupta, R., Oates, S. R., Pandey, S. B., A. J. Castro-Tirado, Jagdish C. Joshi, Y.-D. Hu, A. F. Valeev, B. B. Zhang, **Z. Zhang** et al., GRB 140102A: Insight into Prompt Spectral Evolution and Early Optical Afterglow Emission, *MNRAS*, 505, 4086 (2021)

HONOR & ACHIEVEMENT

- Awarded "Outstanding Presentation" of Galaxy/IGM Workshop at Hirosaki University 08/2024
- Awarded "Excellent graduates" of Guangxi University *06/2019*
- Obtained Second Prize of National College Student Mathematics Competition *11/2018*
- Obtained Winning Prize of National English Competition for College Students *01/2016*
- Obtained Winning Prize in the National College English Contest *11/2016*
- Won the Winning Prize of Guangxi District in the National College Student Mathematical Modeling Contest *01/2017*
- Awarded "Excellent scholarship" of Guangxi University *07/2017*
- Awarded "Excellent scholarship" of Guangxi University 05*/2016*

EXTRACURRICULAR ACTIVITIES

- Planned and hosted the Fun Basketball Games in NJU *06/2019*
- Served as Basketball Team Leader of the School of Astronomy of Nanjing University *09/2021-06/2020*

- Established the First Women's Basketball Team of the School of Physics *09/2017*
- Served as Basketball Team Leader of the School of Physics of Guangxi University *09/2016-06/2018*
- Established the recitation art troupe of the School of Physics *09/2016-06/2017*

TECHNICAL PROFICIENCIES

- Programming Languages: Python, C
- Software: MATLAB, Microsoft Office Suite, Mathematica
- Mathematics & Data Skills: Mathematical modeling, numerical computation, numerical simulation
- Languages: English (fluent), Mandarin Chinese (native), Japanese (intermediate)

Employment History

Research Assistant – Osaka University, Department of Earth and Space Science 10/2022 – 04/2025

- Assisted in numerical simulations of cosmic baryon distribution using CROCODILE simulation.
- Analyzed FRB-related data to study AGN feedback in large-scale structures.
- Co-authored a conference presentation and contributed to manuscript preparation.

Teaching Assistant – The University of Osaka 04/2024 – 08/2024

- Supported instruction in undergraduate "Earth and Space I/II" course.
- Held weekly office hours and graded assignments for 7 students.
- Developed supplementary problem sets and led tutorial sessions.

Tutor – The University of Osaka International Exchange Program 04/2025 – 08/2025

- Provided one-on-one research mentorship to an international exchange student under the Frontier Lab program.
- Supervised data analysis and interpretation for a collaborative astrophysics project. Developed.

Conferences & Workshop

- **Zhang, Z.** (May 2023). *"Probing cosmic baryons with FRBs based on cosmological hydrodynamic simulations."* **Poster** presented at the ILR Kickoff Meeting, Osaka University, Osaka, Japan.
- **Zhang, Z.** (June 2023). *"Magnetar Giant Flares and Intrinsic Spectral Models."* **Contributed Talk** presented at the 8th Academic Seminar of the Guangxi Key Laboratory of Relativistic Astrophysics, Guangxi University, Nanning, China.
- **Zhang, Z.** (April 2024). *"Fast Radio Bursts and Baryon Distribution."* **Contributed Talk** presented at the Keihan Cosmology Meeting, Kyoto-Osaka, Japan.
- **Zhang, Z.** (June 2024). *"Using FRB DM to probe baryon missing problem in the Universe."* **Contributed Talk** presented at LSS QUEST 2024, Osaka University Nakanoshima Center, Osaka, Japan.
- **Zhang, Z.** (August 2024). *"Probing the Baryonic Distribution in Cosmic Large-Scale Structures and Galactic Environments with FRBs based on Hydrodynamic Numerical Simulations."* **Oral presentation** at the Galaxy–IGM Workshop, Osaka, Japan.
- **Zhang, Z.**, Ji, T., Sakai, N. (September 2024). *"Using CNNs to infer the fundamental parameters of the Universe."* **Group project presentation** at AstroAI Workshop, Osaka University, Osaka, Japan. *Role: Core programmer & group presenter.*
- **Zhang, Z.** (March 2025). *"Probing the cosmic baryon distribution and the impact of AGN feedback with FRBs in CROCODILE simulation."* **Contributed Talk** at Early Universe Meeting, University of Tokyo, Komaba Campus, Japan.
- **Zhang, Z.** (March 2025). *"Probing the impact of AGN feedback of massive halos with FRBs in CROCODILE simulation."* **Contributed Talk** at the SPR+ILR Joint Workshop, NAOJ, Mitaka, Japan.
- **Zhang, Z.** (April 2025). *"Probing the cosmic baryon distribution and the impact of AGN feedback with FRBs in CROCODILE simulation."* **Invited speaker** at the Keihan Cosmology Meeting, Kyoto–Osaka, Japan.

- **Zhang, Z.** (2025). *"*Probing the cosmic baryon distribution and the impact of AGN feedback with FRBs in CROCODILE simulation*."* **Contributed Talk** presented at the ILR Kickoff Meeting, The University of Osaka , Osaka, Japan.
- **Zhang, Z.** (May 2025). *"*Using FRBs and the CROCODILE simulation to study the cosmic baryon distribution and the impact of AGN feedback*."* **Contributed Talk** presented at the FRB Conferrence, Beijing Normal University, Beijing, China.
- **Zhang, Z.** (May 2025). *"*Probing the cosmic baryon distribution and AGN feedback impact with FRBs in CROCODILE simulation*."* **Contributed Talk** presented at the Galaxy Workshop, University of Santa Cruz, Santa Cruz, America.
- **Zhang, Z.** (August 2025). *"*Probing the cosmic baryon distribution with FRBs in CROCODILE simulation*"* **Contributed Talk** presented at the FRB @ Cosmology Conferrence, Zhejiang University, Hangzhou, China.
- **Zhang, Z.** (Janaury 2026). "From Cosmic Web to Supernova Remnants: Modeling FRB Dispersion Measures for Baryon Tracing across Scales" **Contributed Talk** presented at the YITP long-term workshop about *Multi-Messenger Astrophysics in the Dynamic Universe*, Yukawa Institute for Theoretical Physics, Kyoto University
- **Zhang, Z.** (February 2026). "Halo Baryon Fraction Analysis based on FRB Cosmology" **Contributed Talk** presented at the DLA40 conference, IPMU, The University of Tokyo.